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A unified model for the deployment of carbon capture and storage

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HIGHLIGHTS

- A mathematical model is developed for carbon capture and storage (CCS) planning.
- The model considers both grid power implications and CO_2 source-sink matching.
- The model determines optimal CO_2 allocation for maximum emissions reduction.
- A realistic case study is solved to illustrate the proposed approach.

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ABSTRACT

This paper presents a comprehensive unified model for planning the retrofit of power plants with carbon capture (CC) technologies and the subsequent carbon dioxide (CO_2) source-sink matching. The planning horizon is divided into time intervals that are not necessarily of equal duration, but which represent time slices generated by specific events (e.g. start and end of plant operation) occurring in the system as well as the required degree of flexibility in planning. In carbon capture and storage (CCS) systems, CO_2 sources have variable flow rates and fixed operating lives, while CO_2 sinks have finite injection rate and storage capacity limits, as well as earliest times of availability. The model takes into account such physical and temporal considerations, and also accounts for the need for additional power generation to compensate for energy loss penalties resulting from the capture of CO_2. A case is used to demonstrate the application of the proposed model. Sensitivity analyses are carried out to examine the tradeoff between carbon emissions reduction and power cost, as well as the effects of uncertainties in sink characteristics and properties of compensatory power on CCS.

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1. Introduction

Carbon capture and storage (CCS) is regarded as one of the most promising approaches to mitigate industrial carbon dioxide (CO_2) emissions on a global scale, other than energy efficiency enhancement and use of low-carbon fuels. CCS involves a two-step approach, beginning with physically or chemically capturing or isolating CO_2 from flue gas, which normally consists of a mixture of atmospheric gases (e.g. nitrogen) and various combustion products [1]. One capture technology is pre-combustion capture in integrated gasification combined cycle (IGCC) plants, wherein CO_2 is separated from raw fuel while the resulting hydrogen-rich stream is used for combustion. Another approach is post-combustion capture using flue gas scrubbing with chemical solvents to separate CO_2 from the combustion products. Alternatively, techniques like oxy-fuel combustion and chemical looping combustion (CLC) burn fuel in the absence of atmospheric nitrogen, and yield flue gas consisting mainly of CO_2 and water vapor. The latter is easily removed via condensation. For any of these capture techniques, the isolated CO_2 is then transported off site, and stored in a sink, which is typically a geological reservoir with sufficient structural integrity to provide essentially permanent storage capability. Examples of possible sinks are saline aquifers, inaccessible coal deposits, and depleted oil or gas reservoirs, provided that geological surveys determine them to be suitable for storage based on geochemical, seismic risk or other physical considerations [1,2]. The capture of CO_2 may also be done in conjunction with increased production of partially depleted reservoirs (i.e. enhanced oil recovery, EOR) or production of a secondary fuel stream from an unviable resource (e.g. enhanced coal bed methane, ECBM, recovery). CCS is thus able to mitigate climate impacts by preventing CO_2 from exiting into the atmosphere.

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Nomenclature

Indices and sets

$i \in \mathcal{I}$	CO_2 sources/power plants
$j \in \mathcal{J}$	CO_2 sinks/storage sites
$k \in \mathcal{K}$	CC technologies
$t \in \mathcal{T}$	time intervals

Parameters

C^{NP}	average cost of electricity from new plants to compensate for energy losses ($/MW y)
C^{UP}	average cost of electricity from unmodified plants ($/MW y)
C_i^U	upper cost rate limit for plant i ($/y)
C_k^{RP}	average cost of electricity from plants retrofitted with CC technology k ($/MW y)
E^{CP}	CO_2 emission factor of compensatory power generation (Mt/MW y)
E_i	baseline CO_2 emission factor of plant i (Mt/MW y)
F_i^U	upper flow rate limit for source i (Mt/y)
F_j	CO_2 injection rate limit of sink j (Mt/y)
H	duration of the planning horizon (y)
L_{ik}	relative energy loss associated with CC technology k in plant i (dimensionless)
M_{ik}	binary parameter indicating the suitability of CC technology k for plant i
P_i	baseline power output of plant i (MW)
Q_j	CO_2 storage capacity of sink j (Mt)
RR_{ik}	carbon removal ratio of CC technology k in plant i (dimensionless)
T_i^{min}	minimum viable duration of CCS for plant i (y)
Y_{it}	binary parameter denoting if plant i operates in time interval t
Y_{jt}	binary parameter denoting the availability of sink j in time interval t
Δ_t	length of time interval t (y)
Γ	limit for the increase in overall electricity cost from the baseline (dimensionless)

Variables

c_{it}	power cost rate of plant i in time interval t ($/y)
f_{ijt}	flow rate of the CO_2 stream from source i to sink j in time interval t (Mt/y)
p_{it}^{loss}	power loss from plant i in time interval t (MW)
y_{ijt}	binary variable denoting the existence of a CO_2 stream from source i to sink j in time interval t
z_{ij}	binary variable indicating the existence of a link between source i and sink j within the planning horizon
z_{ik}	binary variable denoting the decision to use CC technology k in plant i

Although not widely regarded as an inherently cleaner technology, CCS is nonetheless considered an important transitional low-carbon technology [3]. It allows continued use of fossil fuel-fired plants that would otherwise need to be shut down due to carbon emission constraints. This aspect is critical considering the world's continuing dependence on fossil energy [4], which is still relatively reliable and cheap in comparison with most forms of renewable energy. At the same time, fossil energy is regarded as being more socially acceptable than the only other major low-carbon option currently available: nuclear energy [5,6]. The growing demand for affordable electricity in developing countries has also been empirically linked to an increase in demand for coal power [7]. A further development is the recent breakthrough in hydraulic fracturing of non-conventional gas reserves [8], which potentially increases the global natural gas resource to the extent that its use in the rest of the 21st Century will be constrained by CO_2 emissions rather than supply limits. Thus, it appears that in the coming decades, CCS will play a significant role as one of the key "technology wedges" needed to stabilize global greenhouse gas (GHG) emissions to a safe level of about 14 Gt/y [9].

Fossil fuel-fired power plants account for a significant fraction of the world's CO_2 emissions, and are thus some of the prime candidates for CCS. On the other hand, retrofitting power plants for carbon capture (CC) entails major capital costs as well as reduction of both thermal efficiency and power output [10,11]. These penalties ultimately result in an increase in the cost of electricity generation, relative to comparable un-retrofitted power plants. Furthermore, planning the capture of CO_2 requires matching sources with appropriate geological storage sinks, taking into account technical, geographical, geological and economic criteria. Thus, there has been extensive recent work on developing decision aids for planning the commercial deployment of CCS. Some of these works have focused on the power grid implications of CC. Methods based on pinch analysis [12–16] and mathematical programming [17,18] have been used to account for the interplay between CO_2 emissions reduction and power losses that then necessitate compensatory power to be either imported from adjoining regions, or otherwise generated from new (and preferably low-carbon) power plants. Planning of power generation systems taking into account multiple low-carbon options (in addition to CCS) has also been demonstrated by Elkamel et al. [19] and Yang et al. [20]. An interval programming approach has been developed by Zhang et al. [21] to account for various uncertainties in CCS deployment, taking into account the impact of carbon emission trading.

Aside from the effect of retrofits on power generation, the other aspect to be dealt with in CCS planning is the matching of sources and sinks in large-scale CO_2 networks. Roddy [22] outlines some of the main practical considerations in the growth of CO_2 source-sink networks over time, and considers the decision to capture CO_2 as well as assessment of storage options to be two of the key issues that need to be planned carefully. For source-sink matching, mathematical programming approaches may be used to provide vital decision support. For example, Turk et al. [23] proposed an integer linear programming (ILP) model with stream pooling in the context of EOR applications, while Benson and Ogden [24] proposed a similar nonlinear programming (NLP) formulation. Fimbres Weihs et al. [25] proposed nonlinear models for CO_2 network design, and then determined near-optimal solutions to these models using stochastic algorithms. More recently, various mixed integer linear programming (MILP) models have been proposed based on static [26], discrete-time [27–29] and continuous-time formulations [30,31]. Sun and Chen [32] developed an infrastructure model taking into account social impacts associated with topography (e.g. right-of-way issues in populated areas). Based on the model of Tan et al. [30], a robust stochastic MILP formulation has been developed to account for system uncertainties [33]. These MILP models may be readily solved to global optimality using

well-established branch-and-bound methods, although computational effort increases with problem size. Other recent approaches for source-sink matching in CCS systems include rule-based algorithms [34], graphical pinch analysis [35–37] and decision analysis techniques [38,39].

It can be seen that much of the current literature considers the problems of grid power implications and source-sink matching in CCS separately. However, in practice, both issues need to be dealt with simultaneously for effective planning of CCS deployment. The principal contribution of this work is therefore the unification of these two aspects within a single framework. In this paper, a unified multi-period MILP model is developed for planning the retrofit of power plants with CC technologies, as well as the subsequent CO_2 source-sink matching. This model is based on the plant retrofit model of Tan et al. [18] and the discrete-time model of Tan et al. [29] for matching sources and sinks. In addition to physical and temporal considerations, the comprehensive formulation also accounts for the need for additional power generation from new, low-carbon plants to compensate for energy loss penalties of the CC retrofits [18,30]. Alternatively, compensatory power may be drawn from contiguous regions or countries if the geography allows the import of electricity. In either case, additional emissions from compensatory power have to be deducted from the CO_2 reduction benefits resulting from CCS implementation. However, the model presented here is not meant to be a detailed CCS infrastructure planning tool, since such aspects have been dealt with extensively in the literature [25–28,32,34]; thus, constraints on network topology are not considered in this model. An illustrative case study is solved to demonstrate the use of the proposed model to realistic CCS planning problems. Sensitivity analyses are also carried out to illustrate tradeoffs that are bound to occur in real-life deployment of CCS. First, the tradeoff between emission reduction targets and power cost penalties is considered; next, the effects of uncertainties in properties of compensatory power and storage site characteristics are analyzed. The latter has been noted to be a significant factor in planning CCS deployment [2,27,36,40]. Finally, conclusions and prospects for future work are given at the end of the paper.

2. Problem statement

The problem addressed in this paper can be formally stated as follows. The CCS system consists of a set of CO_2 sources $i \in \mathcal{I}$ and a set of CO_2 sinks $j \in \mathcal{J}$. CO_2 sources are power plants to be retrofitted with a set of CC technologies $k \in \mathcal{K}$, while CO_2 sinks are storage sites. Note that the set of CO_2 sources is assumed to be comprised of base load power plants, since peaking plants operating intermittently are not attractive options for CC retrofitting.

- Each CO_2 source has a captured CO_2 flow rate depending on the CC technology used. The start and end of the operating life of each source is predefined.
- Each CO_2 sink has CO_2 injection rate (or injectivity) and storage capacity limits determined by its geological characteristics. The earliest time of availability of each sink is also specified; however, the time at which it ceases to operate (with its capacity limit reached) depends on the rate of CO_2 injection, and thus in general cannot be specified *a priori*. In other words, a given sink operates either until the end of the planning horizon, or until it is filled to capacity.
- Each CC technology is characterized by fixed CO_2 removal and energy loss ratios as well as a fixed relative or dimensionless cost. The latter parameter gives the ratio of cost of power generated by the retrofitted plant to the unmodified plant. Cost ratios are assumed to be exogenously defined, so that the model does not account for technological learning internally [41].

Since the operating lives of CO_2 sources and sinks may not completely coincide, source-sink matches need to be made during periods of overlap. To address such temporal issues, the planning horizon is divided into a number of time intervals $t \in \mathcal{T}$, with events taking place only at interval boundaries. The objective is to maximize CO_2 emissions reduction by retrofitting power plants and matching CO_2 sources and sinks, while keeping the increase in electricity cost within a specified limit and satisfying all temporal and physical constraints.

3. Model formulation

In this work, a mathematical programming model is developed for unified planning of CCS deployment based on a superstructure shown in Fig. 1, considering both grid implications and source-sink matching. This model requires that data be specified for CO_2 sources and sinks, CC technologies and compensatory power generation. With these input data, the model is able to determine an optimal CO_2 allocation network maximizing the amount of CO_2 captured and stored within the given system. In addition, the model developed here assumes a scenario where any demand growth is factored implicitly in the entry of new power plants during the planning horizon. The mathematical formulation is as follows; notation used is given in the Nomenclature.

For each power plant to be retrofitted, only one CC technology can be chosen as given in the following equation:

$$\sum_{k\in\mathcal{K}} z_{ik} \leqslant 1 \quad \forall i \in \mathcal{I} \tag{1}$$

To account for possible incompatibilities between specific power generation and CC technologies, Eq. (2) is introduced to define the forbidden ($M_{ik} = 0$) and allowable matches ($M_{ik} = 1$):

$$z_{ik} \leqslant M_{ik} \quad \forall i \in \mathcal{I}, k \in \mathcal{K} \tag{2}$$

It is assumed that each CO_2 source may be connected to only one sink, as given in Eq. (3); however, a sink can be linked to multiple sources.

$$\sum_{j\in\mathcal{J}} z_{ij} \leqslant 1 \quad \forall i \in \mathcal{I} \tag{3}$$

Eq. (4) ensures that a CC-retrofitted power plant is matched with a sink for CO_2 storage. On the other hand, no such arrangements will be made for non-CC plants.

$$\sum_{j\in\mathcal{J}} z_{ij} = \sum_{k\in\mathcal{K}} z_{ik} \quad \forall i \in \mathcal{I} \tag{4}$$

A link between a source and a sink should only be made if its duration makes it worthwhile. This constraint is set by the following equation:

$$T_i^{\min} z_{ij} \leqslant \sum_{t\in\mathcal{T}} y_{ijt}\Delta_t \leqslant H z_{ij} \quad \forall i \in \mathcal{I}, j \in \mathcal{J} \tag{5}$$

Once a link is established, it is assumed to remain in place with a constant flow of CO_2 until the end of the source's operating life:

$$y_{ij,t-1} - y_{ijt} \leqslant 1 - Y_{it} \quad \forall i \in \mathcal{I}, j \in \mathcal{J}, t \in \mathcal{T}, t > 1 \tag{6}$$

Note that Eq. (6) is inactive when plant i is not operating ($Y_{it} = 0$), in which case Eq. (7) ensures no CCS for the plant:

$$y_{ijt} \leqslant Y_{it} \quad \forall i \in \mathcal{I}, j \in \mathcal{J}, t \in \mathcal{T} \tag{7}$$

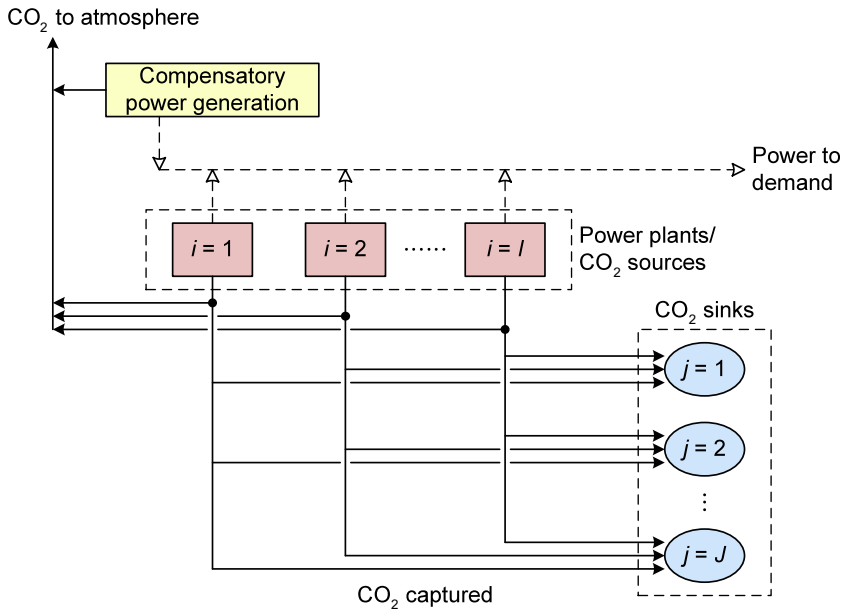


Fig. 1. Superstructure for a CCS system.

The captured CO_2 flow rate can be calculated using Eqs. (8) and (9). When plant i is sending CO_2 to sink j ($y_{ijt} = 1$), the flow rate is given by Eq. (8); for non-existing CO_2 streams ($y_{ijt} = 0$), Eq. (9) ensures zero flow rates.

$$F_i^{U}(y_{ijt} - 1) \leqslant f_{ijt} - P_i E_i \sum_{k\in\mathcal{K}} RR_{ik} z_{ik} \leqslant 0 \quad \forall i \in \mathcal{I}, j \in \mathcal{J}, t \in \mathcal{T} \tag{8}$$

$$f_{ijt} \leqslant F_i^{U} y_{ijt} \quad \forall i \in \mathcal{I}, j \in \mathcal{J}, t \in \mathcal{T} \tag{9}$$

The CO_2 injection rate and the amount of CO_2 to be stored in a sink are restricted by Eqs. (10) and (11) respectively:

$$\sum_{i\in\mathcal{I}} f_{ijt} \leqslant F_j Y_{jt} \quad \forall j \in \mathcal{J}, t \in \mathcal{T} \tag{10}$$

$$\sum_{i\in\mathcal{I}} \sum_{t\in\mathcal{T}} f_{ijt} \Delta_t \leqslant Q_j \quad \forall j \in \mathcal{J} \tag{11}$$

Eqs. (12) and (13) are used to determine the CC power loss. Note that $\Sigma_j y_{ijt}$ is indirectly limited by Eqs. (3) and (5) to a maximum of one. When plant i is having CCS (i.e. $\Sigma_j y_{ijt} = 1$), the resulting power loss is given by Eq. (12). Otherwise, Eq. (13) ensures no power losses.

$$P_i \left(\sum_{j\in\mathcal{J}} y_{ijt} - 1 \right) \leqslant p_{it}^{loss} - P_i \sum_{k\in\mathcal{K}} L_{ik} z_{ik} \leqslant 0 \quad \forall i \in \mathcal{I}, t \in \mathcal{T} \tag{12}$$

$$p_{it}^{loss} \leqslant P_i \sum_{j\in\mathcal{J}} y_{ijt} \quad \forall i \in \mathcal{I}, t \in \mathcal{T} \tag{13}$$

For economic considerations, the power generation cost can be calculated using Eqs. (14)–(16). The first two constraints apply when plant i operates without ($Y_{it} = 1$; $\Sigma_j y_{ijt} = 0$) and with CCS ($\Sigma_j y_{ijt} = 1$), respectively; the last one ensures zero power cost outside the operating life of plant i ($Y_{it} = 0$).

$$C_i^{U} \left(Y_{it} - \sum_{j\in\mathcal{J}} y_{ijt} - 1 \right) \leqslant c_{it} - C^{UP} P_i \leqslant C_i^{U} \left(1 - Y_{it} + \sum_{j\in\mathcal{J}} y_{ijt} \right) \quad \forall i \in \mathcal{I}, t \in \mathcal{T} \tag{14}$$

$$C_i^{U} \left(\sum_{j\in\mathcal{J}} y_{ijt} - 1 \right) \leqslant c_{it} - \sum_{k\in\mathcal{K}} C_k^{RP} P_i (1 - L_{ik}) z_{ik} \leqslant C_i^{U} \left(1 - \sum_{j\in\mathcal{J}} y_{ijt} \right) \quad \forall i \in \mathcal{I}, t \in \mathcal{T} \tag{15}$$

$$c_{it} \leqslant C_i^{U} Y_{it} \quad \forall i \in \mathcal{I}, t \in \mathcal{T} \tag{16}$$

Eq. (17) limits the additional cost of electricity to the system:

$$\sum_{i\in\mathcal{I}} \sum_{t\in\mathcal{T}} c_{it} \Delta_t + C^{NP} \sum_{i\in\mathcal{I}} \sum_{t\in\mathcal{T}} p_{it}^{loss} \Delta_t \leqslant \Gamma C^{UP} \sum_{i\in\mathcal{I}} \sum_{t\in\mathcal{T}} P_i Y_{it} \Delta_t \tag{17}$$

The objective function is to maximize the reduction of total carbon emissions of the system during the entire planning horizon:

$$\max_{\mathbf{x}\in\mathbf{\Omega}} \phi = \sum_{i\in\mathcal{I}} \sum_{j\in\mathcal{J}} \sum_{t\in\mathcal{T}} f_{ijt} \Delta_t - E^{CP} \sum_{i\in\mathcal{I}} \sum_{t\in\mathcal{T}} p_{it}^{loss} \Delta_t \tag{18}$$

$$\mathbf{x} \equiv \left\{ \begin{array}{c} c_{it}, f_{ijt}, p_{it}^{loss}, y_{ijt}, z_{ij}, z_{ik} \\ \forall i \in \mathcal{I}, j \in \mathcal{J}, k \in \mathcal{K}, t \in \mathcal{T} \end{array} \right\}$$

$$\Omega = \{\mathbf{x} | \text{Equations(1)-(17)}\} \tag{20}$$

where $\mathbf{x}$ is a vector of the variables, and $\mathbf{\Omega}$ a feasible solution space defined by the constraints. The first term on the right hand side of Eq. (18) gives the total reduction in emissions from plants retrofitted for CCS, while the second term gives the additional emissions from new power plants needed to compensate for CC energy losses. Thus, the difference between these two terms gives the net emissions reduction relative to the baseline grid. Note that the model is an MILP, which can be readily solved to global optimality without significant computational difficulties.

In the next section, the use of the model is demonstrated through a hypothetical but realistic case study. The model is implemented in the GAMS environment [42] on a Core 2, 2.00 GHz processor with CPLEX as the MILP solver. Solutions were found with negligible processing time (<10 CPU s) for most cases.

Table 1
CO_2 source data for the case study.

Plant/source	Fuel	Capacity, P_i (MW)	Emission factor, E_i (Mt/MW y)	Time of operation (y)	Emissions (Mt)
1	Coal	400	0.01	0–30	120
2	Coal	500	0.01	0–40	200
3	Coal	250	0.01	5–30	62.5
4	Coal	800	0.01	10–40	240
5	Natural gas	1000	0.005	0–40	200
6	Natural gas	600	0.005	10–50	120

Table 2
CO_2 sink data for the case study.

Sink	Injection limit, F_j (Mt/y)	Start time (y)	Maximum storage, Q_j (Mt)
A	5	0	200
B	10	10	400
C	10	15	250

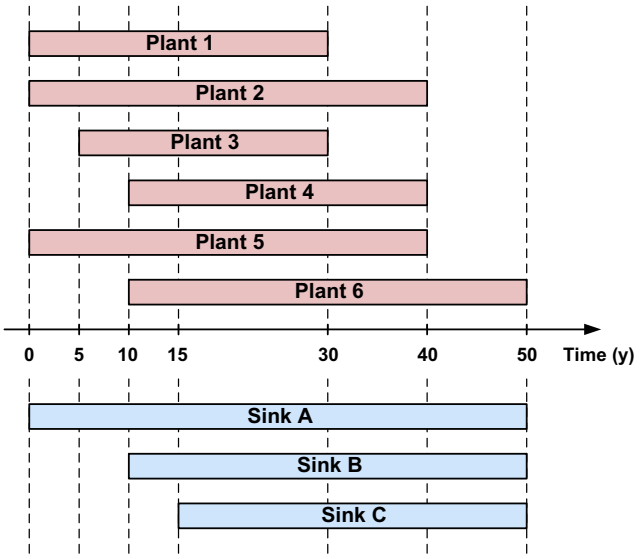


Fig. 2. Gantt chart for the case study.

4. Case study

The illustrative case study, adapted from Tan et al. [18,30], involves planning for CCS over a 50-y time horizon given six power plants as CO_2 sources and three geological reservoirs as CO_2 sinks, which are assumed to be located in close proximity. The source and sink data are shown in Tables 1 and 2 respectively. Fig. 2 shows the corresponding Gantt chart. Note that the plant operating lives and the sink start times divide the planning horizon into six time intervals. However, for flexible CCS planning, more time intervals should be defined before the latest allowable time for plants to begin CO_2 capture. This is because CCS can commence only at the start of a time interval. In this case study, the minimum viable duration of CCS for each plant is assumed to be 20 years; hence, the planning period before the 30th year (i.e. 0–30 y) is further divided into fine intervals of one year each, with the division of the period 30–50 y remaining the same. Thus, 32 time intervals are used in total.

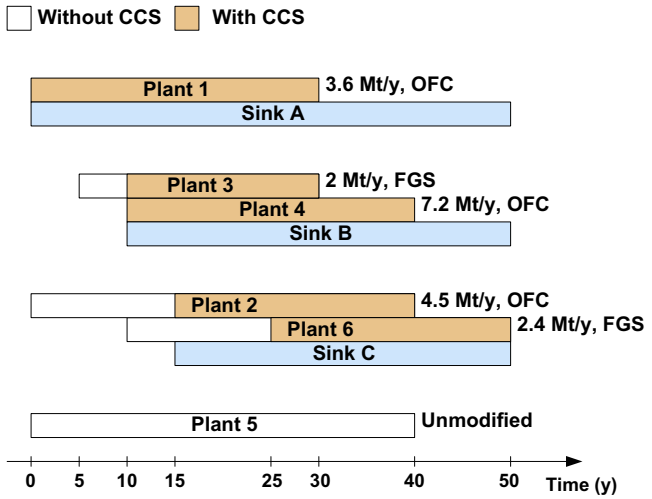


Fig. 3. Gantt chart representation of the optimal source-sink matching ($\Gamma = 1.3$).

Two CC retrofits are considered in this case study for GHG emissions reduction, namely flue gas scrubbing (FGS) and oxy-fuel combustion (OFC). Plants 3 and 6 are unsuitable for the OFC option; for the rest, there are no other technological compatibility restrictions. For both coal and gas plants, the carbon removal ratios of FGS and OFC are as assumed to be 0.8 and 0.9 respectively, and the relative power losses are 0.2 and 0.25 respectively. In addition, the emission factor of compensatory power is taken as 0.001 Mt/MW y, assuming that the power mix is dominated by low-carbon sources (e.g. wind and solar energy).

It is also assumed that each unit of electricity produced in the retrofitted plants is 60% more expensive than that produced by the unmodified plants; likewise each unit generated by the new plants costs 40% more than the baseline cost. Hence, the cost

Table 3
Optimal source-sink matching for $\Gamma = 1.3$.

Source	Sink A	Sink B	Sink C	Source summary
1	3.6 Mt/y (0–30 y)	0	0	90% of available CO_2 (120 Mt) captured
2	0	0	4.5 Mt/y (15–40 y)	56.25% of available CO_2 (200 Mt) captured
3	0	2 Mt/y (10–30 y)	0	64% of available CO_2 (62.5 Mt) captured
4	0	7.2 Mt/y (10–40 y)	0	90% of available CO_2 (240 Mt) captured
5	0	0	0	0% of available CO_2 (200 Mt) captured
6	0	0	2.4 Mt/y (25–50 y)	50% of available CO_2 (120 Mt) captured
Sink summary	108 Mt (54%) of storage capacity used	256 Mt (64%) of storage capacity used	172.5 Mt (69%) of storage capacity used	536.5 Mt of CO_2 emissions captured and stored (56.92% of available CO_2 and 63.12% of storage capacity)

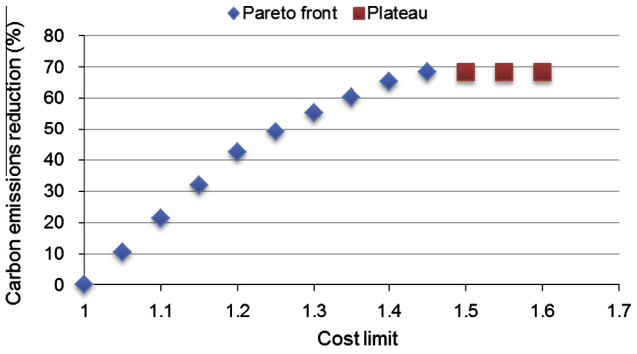


Fig. 4. Carbon reduction and dimensionless cost tradeoff.

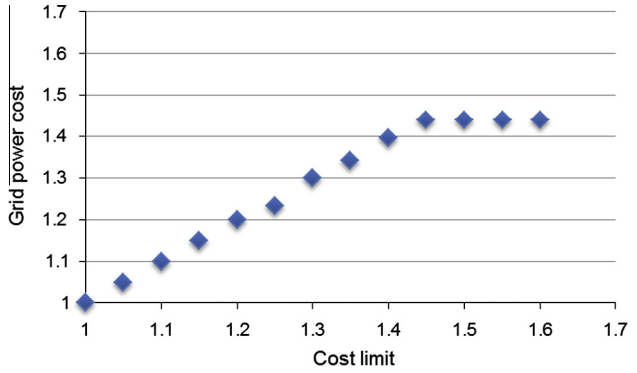


Fig. 5. Actual dimensionless grid power cost versus the dimensionless cost limit.

parameters (C^{UP}, $C^{RP}_{k=FGS}/C^{RP}_{k=OFC}$ and C^{NP}) used for Eqs. 14, 15, and 17 can be given in dimensionless values (relative to unmodified plants) as 1, 1.6/1.6 and 1.4, respectively; such use of dimensionless cost coefficients was previously proposed by Tan et al. [18]. The objective is to maximize carbon emissions reduction without increasing the overall electricity cost by more than 30%, i.e. $\Gamma = 1.3$.

Solving the MILP model (with 3977 constraints, 1059 continuous variables and 508 binary variables) yields an optimal result requiring the retrofit of plants 1 (OFC), 2 (OFC), 3 (FGS), 4 (OFC) and 6 (FGS), with plant 5 remaining unmodified. The resulting electricity mix is exactly 30% more expensive than the original power prior to CC retrofitting. Table 3 shows the optimal source-sink matching. Alternatively, this CO_2 network may be represented on a Gantt chart as shown in Fig. 3. Plant 1 is linked to sink A, plants 3 and 4 are linked to sink B, while plants 2 and 6 are connected to sink C. This allocation results in the capture and storage of 536.5 Mt of CO_2, which accounts for 56.92% of the available CO_2 for capture (942.5 Mt) and 63.12% of the total storage capacity (850 Mt). Note that plants 2 and 3 begin to send CO_2 for storage as soon as their respective sinks become available. These matches

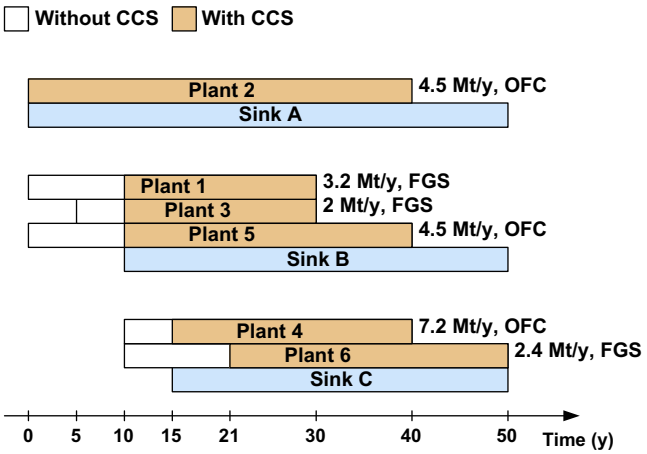


Fig. 6. Gantt chart representation of the optimal source-sink matching ($\Gamma \geqslant 1.45$).

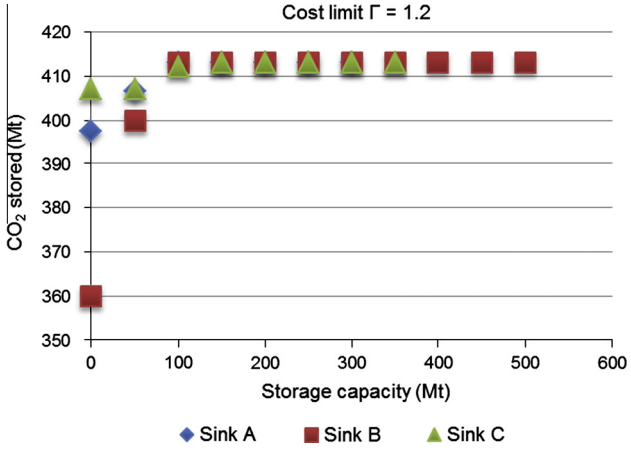


Fig. 7. Effects of changes in sink capacity ($\Gamma = 1.2$).

are constrained by temporal considerations (i.e. sink availability). By contrast, plant 6 does not begin CCS until the 25th year because of the cost limitation.

Furthermore, the objective function value is only 520.375 Mt, due to the penalty of 16.125 Mt from new plants needed to compensate for power losses amounting to 16,125 MW y through the planning horizon. This drop in output thus results in additional CO_2 emissions that need to be deducted from the carbon reduction benefits of CCS. The net reduction is 55.21% from the baseline level.

As the cost limit for power generation increases, it may be expected that more extensive retrofitting becomes possible; hence, it is essential to explore the tradeoff between economic targets and environmental benefits. The analysis of the sensitivity of the total carbon emissions reduction of the entire system to the cost

Table 4
Optimal source-sink matching for $\Gamma \geqslant 1.45$.

Source	Sink A	Sink B	Sink C	Source summary
1	0	3.2 Mt/y (10–30 y)	0	53.33% of available CO_2 (120 Mt) captured
2	4.5 Mt/y (0–40 y)	0	0	90% of available CO_2 (200 Mt) captured
3	0	2 Mt/y (10–30 y)	0	64% of available CO_2 (62.5 Mt) captured
4	0	0	7.2 Mt/y (15–40 y)	75% of available CO_2 (240 Mt) captured
5	0	4.5 Mt/y (10–40 y)	0	67.5% of available CO_2 (200 Mt) captured
6	0	0	2.4 Mt/y (21–50 y)	58% of available CO_2 (120 Mt) captured
Sink summary	180 Mt (90%) of storage capacity used	239 Mt (59.75%) of storage capacity used	249.6 Mt (99.84%) of storage capacity used	668.6 Mt of CO_2 emissions captured and stored (70.94% of available CO_2 and 78.66% of storage capacity)

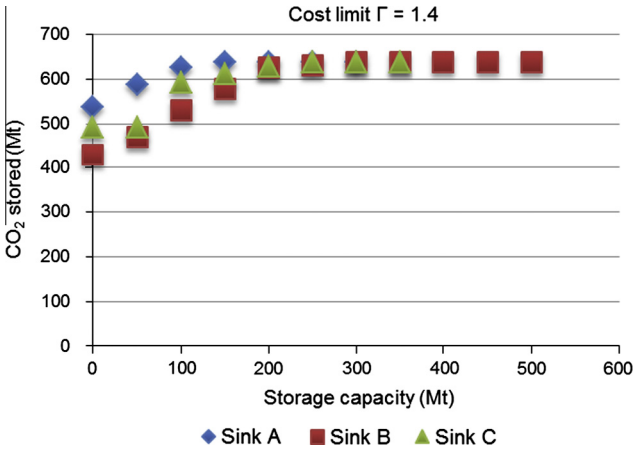


Fig. 8. Effects of changes in sink capacity ($\Gamma = 1.4$).

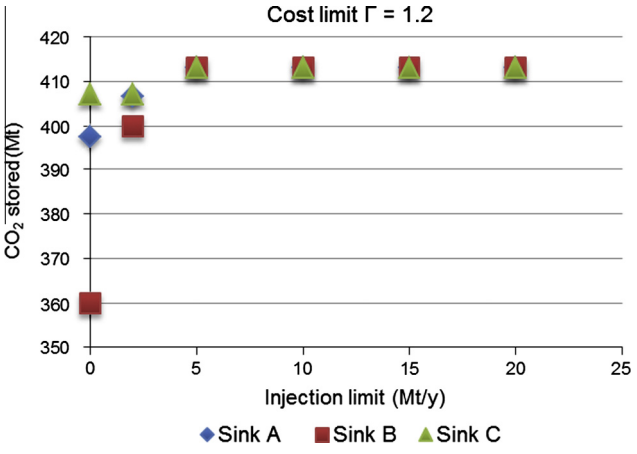


Fig. 10. Effects of changes in sink injectivity ($\Gamma = 1.2$).

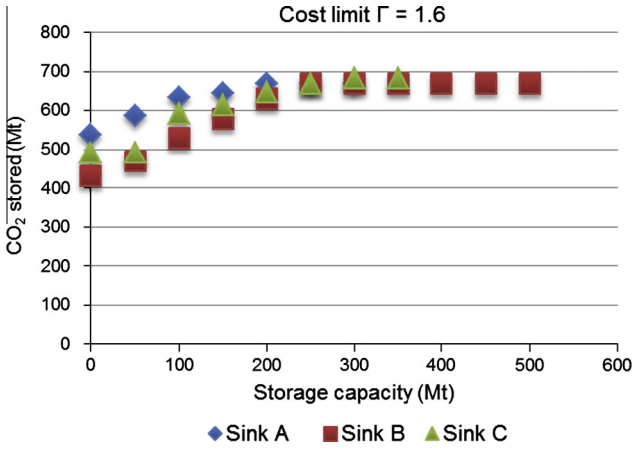


Fig. 9. Effects of changes in sink capacity ($\Gamma = 1.6$).

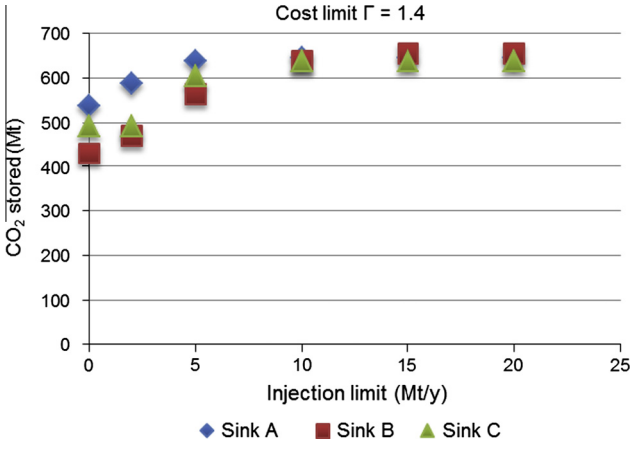


Fig. 11. Effects of changes in sink injectivity ($\Gamma = 1.4$).

limit is shown in Fig. 4, where the diamond points represent the frontier of Pareto optimal solutions. It can be seen that the reduction in carbon emissions at first increases in a straight line as the cost limit is relaxed in the range of 1–1.25; then, the rate of improvement begins to decline, as further costs give diminishing returns. The benefit levels off at 68.44% carbon emissions reduction for dimensionless power cost limits of 1.45 and above, with the actual grid power cost also leveling off at 1.44 (see Fig. 5), i.e. 44% more expensive than the baseline cost. The optimal results for $\Gamma \geqslant 1.45$ are shown in Table 4 and Fig. 6. This solution is to retrofit plants 1, 3 and 6 with FGS, and plants 2, 4 and 5 with OFC. Note that plant 1 can only use FGS, which gives a lower CO_2 flow rate, in order not to exceed the injection limit of sink B. It can also be seen that plant 6 begins to send CO_2 to sink C only at the start of the 21st year because of the capacity limit of the sink. With the cost limit relaxed, storage constraints become apparent, which in turn restrict the extent of CCS. Hence, the power cost limit eventually becomes non-binding.

The previous results clearly illustrate the tradeoff between the reduction in the emission level and the increase in electricity generation cost. At the same time, there may be uncertainties in the model parameters due to significant uncertainty inherent in the characteristics of geological reservoirs. The planning of CCS systems assumes that the geological survey of the sink determines its storage capacity. However, in practice, estimates for any given site may vary depending on the survey technique used [2,43]. In addition, various physical factors such as depth and temperature can affect the true storage capacity of saline aquifers [44] and coal seams [45]. Further surveys may of course be conducted to obtain more precise estimates of a sink's storage capacity, but only at added cost and delay. Recent research suggests that storage capacity is one of the key determinants of the extent of benefits to accrue from CCS deployment [46,47]. The results of sensitivity analysis for sink capacity for $\Gamma = 1.2$, 1.4 and 1.6 are shown in Figs. 7–9 respectively. In the former cases, it can be seen that increases in storage capacity have no effect on the amount of CO_2 to be stored, as the main limiting factor is the cost limit itself. However, decreases in capacity can further restrict the opportunities for CCS. In the case of $\Gamma = 1.6$, only the capacity increase of sink C can increase the storage of CO_2; while the increases of sinks A and B again have no effect, because CO_2 storage is also restricted by their injection limits.

Other than sink capacity, the injection rate limit (injectivity) of a sink is also an important parameter in CCS planning. The injectivity of a geological sink is determined by factors including the inherent physical properties (i.e. porosity and permeability) of the underground rock strata [36]. In addition, injectivity can be controlled by engineering interventions – e.g. increasing the injectivity by adding injection wells at the storage site. Due to sampling error, further geological surveys of a site may determine that the sinks possess different properties from a previous estimate. Moreover, the inherent injectivity of geological sinks may change over time through chemical reactions due to the injection of CO_2. The

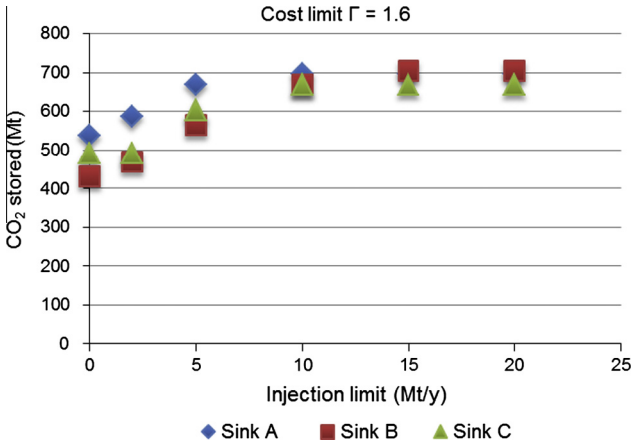


Fig. 12. Effects of changes in sink injectivity (Γ = 1.6).

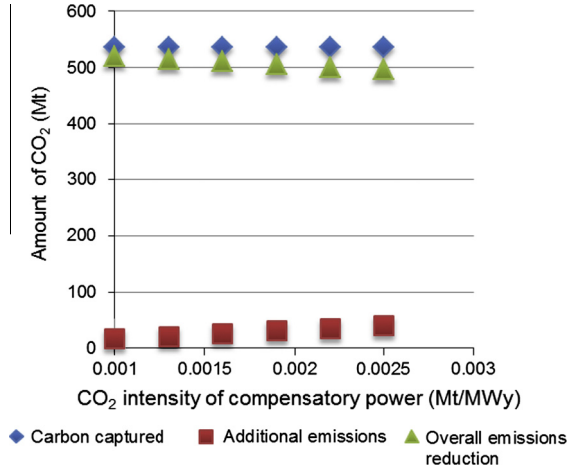


Fig. 13. Effects of increase in the CO_2 intensity of compensatory power (Γ = 1.3).

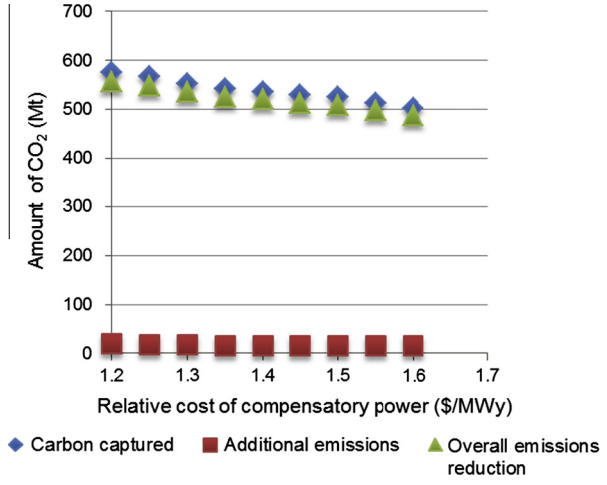


Fig. 14. Effects of changes in the compensatory power cost (Γ = 1.3).

results of sensitivity analysis for sink injectivity for Γ = 1.2, 1.4 and 1.6 are shown in Figs. 10–12 respectively. Similar to the case of sink capacity, increases in injectivity have no effect on CCS for low cost limits (i.e. Γ = 1.2), but injectivity decreases can further restrict the extent of CCS. For higher cost limits (Γ = 1.4 and 1.6), both the injectivity increases of sinks A and B can increase the amount of CO_2 stored. The increase of sink C still has no effect because of its storage capacity limit.

It has been noted that there still remains considerable uncertainty in predicting the economic cost as well as life cycle-based carbon intensity of CCS installations [48]. Part of the uncertainty in the life cycle carbon footprint arises from the difficulty of predefining the technology to be used to compensate for energy losses resulting from CC retrofits. In this work, the effects of changes in the emission factor and the cost of compensatory power on CCS are analyzed to assess the robustness of the optimal results. As shown in Fig. 13, the amount of CO_2 to be captured remains unchanged as the CO_2 intensity of compensatory power increases; therefore, the overall emissions reduction will decrease with the increase in additional emissions. Although not found in the analysis, it might be the case for the system to reduce the extent of possible CCS to avoid additional emissions from degrading the overall carbon reduction benefits, when the CO_2 intensity of compensatory power becomes even higher. Fig. 14 shows that the extent of CCS decreases with the increase in compensatory power cost.

5. Conclusion

A comprehensive model for CCS deployment has been developed in this paper. This model unifies the problems of grid effects of power plant retrofitting and the subsequent CO_2 source-sink matching. The need for additional power generation from new, low-carbon plants to compensate for CC energy losses is also accounted for. An illustrative case study was solved to demonstrate the use of the model to realistic CCS planning problems. Sensitivity analysis has been performed to show the interplay between carbon emissions and power losses in the system. Further analyses were also done to assess the effects of parametric uncertainties. In future work, the model will be expanded to take into account detailed schedule changes and electricity demand growth. Geographical aspects, network topology and capital cost considerations may also be integrated in future variants. Finally, effects of technological learning on power costs may be integrated to account for the long time horizon involved.

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